

Programme	B. Sc. Computer Science				
Course Title	Internet of Things				
Type of Course	Major				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Understanding of Computer Networks, Hardware and Sensors. 2. Basic Programming Skills. 3. Understanding of basic Internet Terminologies				
Course Summary	The course covers a comprehensive overview of Internet of Things (IoT) concepts, technologies, and architectures. It begins by explaining the foundational framework and architectural views of IoT systems, highlighting machine-to-machine (M2M) communication and various IoT sources. Design principles for connected devices are emphasized, focusing on IoT system layers and standardization. Communication technologies, data enrichment, and device management at gateways are explored in detail. The importance of internet connectivity principles, web connectivity design, and protocols are discussed. Additionally, the syllabus delves into message communication protocols and networking using gateways. Sensor technology, participatory sensing, and industrial and automotive IoT applications are also covered. Furthermore, it includes embedded computing basics, platforms like Arduino and Raspberry Pi. Finally, cloud Platform as a Service (PaaS) as well as essential networking technologies are introduced.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding the concepts and architecture of IoT involves grasping the fundamental principles and interconnected structures of its diverse components.	U	F	Instructor-created exams / Quiz
CO2	Understanding the hardware components of IoT involves recognizing sensors, actuators, communication modules, and processing units, crucial for data collection, transmission, and analysis.	U	C	Practical Assignment / Observation of Practical Skills
CO3	Explain the design principles for connected devices, focusing on IoT system layers and standardization. Communication technologies, data enrichment, and device management at gateways	U	C	Practical Assignment / Observation of Practical Skills

CO4	Perceive the basic protocols in IoT, that enable efficient communication between devices, facilitating data exchange and interoperability within IoT networks.	Ap	C	Instructor-created exams / Quiz
CO5	Demonstrating IoT utilizes popular hardware and software platforms showcasing practical implementations of interconnected devices.	Ap	P	Practical Assignment / Observation of Practical Skills
CO6	Implementing IoT in real-time situations by deploying interconnected devices to collect, process, act upon data and visualize them..	Ap	P	Practical Assignment / Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Internet of Things: An Overview		9	15
	1	Internet of Things (IoT) - Conceptual Framework, Architectural View	2	
	2	IoT Technology, IoT Sources, M2M Communication, Examples of IoT	2	
	3	Design Principles for Connected Devices - Introduction , IoT/M2M System Layers and Design Standardisation	2	
	4	Communication Technologies, Data Enrichment	1	
	5	Data Consolidation and Device Management at Gateway	2	
II	Internet Connectivity Principles		9	18
	6	Internet Connectivity, Internet-based Communications, IP Addressing in the IOT	2	
	7	Media Access Control, Application Layer Protocols, HTTP, HTTPS, FTP	2	
	8	Web Communication protocols for Connected Devices	1	
	9	Message Communication Protocols for Connected Devices	1	
	10	Network using Gateway	1	

	11	SOAP, REST, HTTP RESTful and Web Sockets	2	
III	Sensors, Participatory Sensing, RFIDs, and Wireless Sensor Networks		12	17
	12	Sensor Technology, Participatory Sensing	3	
	13	Industrial IoT, Automotive IoT	3	
	14	Actuators, Sensor Data Communication Protocols	3	
	15	RF Identification Technology, Wireless Sensor Networks Basics	3	
IV	Basics of Embedded Computing		15	20
	16	Introduction, Embedded Hardware Unit	2	
	17	Basics of Embedded Platforms for Prototyping - Arduino, Raspberry Pi	2	
	18	Basics of Embedded Platforms for Prototyping - Intel Galileo, Intel Edison, BeagleBone, mBed	2	
	19	Prototyping Embedded Device Software for IoT using Arduino	3	
	20	Understanding Cloud PaaS - Xively, Nimbits, IBM Bluemix, CISCO IoT, AWS IoT, TCS Connected AWS Platform	2	
	21	Basics of popular technologies - IEEE 802.15.4, Zigbee Specification, WiFi, 6LoWPAN, IPv6, LoRaWAN	2	
	22	IoT - Vulnerabilities, Security and Threats	2	
V	Practical Activities		30	

	Following are some of the suggested practical activities. 1. Temperature and Humidity Monitoring System: Use sensors to measure temperature and humidity. Connect the sensor to Arduino or Raspberry Pi and send the data to a cloud platform . Visualize the data in real-time on a web dashboard. 2. Smart Home Automation: Control home appliances such as lights, fans etc. using Arduino or Raspberry Pi. Interface relay modules or solid-state relays with the microcontroller to control the appliances remotely. Use MQTT protocol for communication and control the devices via a mobile app or web interface. 3. Weather Station: Build a weather station using sensors like BMP180 or BME280 for measuring temperature, pressure, and altitude. Interface the sensors with Arduino or Raspberry Pi and log the data to a cloud platform. Display weather data on an LCD screen or visualize it on a web dashboard. 4. Smart Plant Monitoring System: Monitor the soil moisture level, light intensity, and temperature to create a smart plant monitoring system. Use sensors like soil moisture sensors, LDRs, and temperature sensors with Arduino or Raspberry Pi. Send notifications to users when plants need watering or when light conditions are inadequate. 5. Home Security Camera System: Build a home security camera system using Raspberry Pi and a USB webcam or Raspberry Pi Camera Module. Stream live video footage over the network and access it remotely using a web browser or mobile app. Implement motion detection algorithms to trigger recording or notifications.	30	
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Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	3	-	-	-	-						
CO 2	1	3	-	-	-	-						
CO 3	1	3	-	-	-	-						
CO 4	1	2	-	-	-	-						
CO 5	1	2	3	3	-	3						
CO 6	1	2	3	3	-	3						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓	✓	✓
CO 3	✓	✓	✓	✓
CO 4		✓	✓	✓
CO 5		✓		✓
CO 6	✓	✓	✓	✓

References:

1. Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill (India) Private Limited.
2. Arshadeep Bahga, Vijay Madisetti, “Internet of Things: A hands-on approach”, University Press, 2015 (First edition)
3. Dieter Uckelmann, Mark Harrison, Michahelles Florian (Ed.), Architecting the internet of things, Springer, 2011
4. Dr. Ovidiu Vermesan, Dr. Peter Friess, Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, River Publishers, 2013
5. C. Doukas, "Building Internet of Things with the Arduino," Packt Publishing, 2016.
6. Dr. Ovidiu Vermesan, Dr. Peter Friess, Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, River Publishers, 2013
7. Simon Monk, “Programming Arduino: Getting Started with Sketches”, McGraw Hill Publications